Exam Date & Time: 04-Jan-2025 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

B.Tech VII Semester MAKEUP EXAMINATION DEC/JAN 2024/25
PROGRAM ELECTIVE III (COMMON TO ALL BRANCHES)

COMPUTATIONAL PROBABILITY AND DESIGN OF EXPERIMENTS [MAT 4053]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions. Section Duration: 180 mins

Answer all questions

1A) Based on the data on life times in days of bulbs, obtain the maximum likelihood estimate of the parameter θ of the distribution $f(x;\theta) = \begin{cases} \frac{1}{\theta}, 0 < x < \theta \\ 0, otherwise \end{cases}, \theta > 0$ (3)

22.0, 23.9, 20.9, 23.8, 25.0, 24.0, 21.7, 23.8, 22.8, 23.1, 23.1, 23.5, 23.0, 23.0

1B) A random sample of ten students had the following scores in Mathematics out of 150.

70, 120, 110, 101, 88, 83, 95, 98, 107, 100.

Do these data support the assumption of a population average of 100? (3)

At 5% level of significance, $t(9) = 2.26, t(10) = 2.23$

1C) The data below is obtained from a random sample of 300 college students. The students were classified with respect to size of the high school from which they were graduated and with respect to their freshman year grade point average. Test whether there is any relationship between these two classified variables. At 5% level of significance, $\chi^2(2) = 5.99, \chi^2(3) = 7.81$ (4)

Record	High School		
	Small	Medium	Large
> 2.5 average	18	51	46
< 2.5 average	42	79	64

2A) Write the incidence matrix for the design below having seven treatments and seven (3)

Blocks	Treatments		
1	1	2	4
2	2	3	5
3	3	4	6
4	4	5	7
5	5	6	1
6	6	7	2
7	7	1	3

blocks.

2B) Identify the design and estimate the missing observation in the layout. Write the degrees of freedom corresponding to the random cause of variation.

A 12	**C** 13	**B** 10	**D** 15
C 18	**B** 12	**D** 16	**A**
B 14	**D** 10	**A**15	**C**21
D 12	**A** 17	**C** 19	**B** 21

(3)

2C) Find the main effects and interaction effects totals in the following experiment using the Yates' table.

Blocks				
I	(1) 23	k 25	p 22	kp 38
II	p 40	(1) 26	k 36	kp 38
III	(1) 29	k 20	kp 30	p 20
IV	kp 34	k 31	p 24	(1)28

(4)

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3A) The following is the association scheme of a particular design with 10 treatments labelled 1,2, …,10. Identify the association type specifying the design.

*	1	2	3	4
1	*	5	6	7
2	5	*	8	9
3	6	8	*	10
4	7	9	10	*

(3)

3B) Three different makes of diesel engines were tested to determine the useful lifetimes before an overhaul was needed. If the engine life span for each make are normally distributed and have the same variance, test at 5% level of significance, whether the average life of the three makes are the same. At 5% level of significance, F(2, 12) = 3.89, F(2, 14) = 3.74, F(12, 14) = 2.53

(3)

A	6	2	4	1	7
B	8	7	7	2	6
C	3	2	5	4	1

3C) The following table gives the yield (pounds per plot) of three varieties of wheat obtained with four different kinds of fertilizers. Stating the necessary assumptions and the required model, test the hypothesis of equality in the average yields of the three varieties of wheat and the null hypothesis that the four fertilizers are equally effective. $F_{(2,6)}(0.05) = 5.14, F_{(3,6)}(0.05) = 4.76$

Fertilizer	Variety of Wheat		
	A	B	C
α	6	4	8
β	9	3	4
γ	4	5	3
δ	5	4	7

(4)

4A) The objective of an experiment was to determine if there is a significant difference in the average yield of rice when three different fertilisers were used. Based on the data collected, the following operations were performed. Mention the reasons for performing the same.

i. Box plot

ii. Shapiro-Wilk's test

iii. Levene's test

(3)

4B) Based on the ANOVA and Tukey's HSD given below, write your inference considering the level of significance as 5%.

Source of Variation	Df	SS	MSS	F	p-value
Fertilizer	2	6.07	3.03	7.86	0.0000
Error	93	35.80	0.3859		

Pairs	Difference	Confidence Interval	p-value
2 - 1	0.18	(-0.19, 0.55)	0.4954
3 - 1	0.62	(0.23, 0.97)	0.0006
3 - 2	0.42	(0.05, 0.79)	0.0209

(3)

4C) The following observations are drawn sequentially from $N(\theta, 1)$.

-0.65, -1.2, 1.7, -1.4, -0.6, 1, 0, 0, 1.1, 1.6

Test the hypothesis $H_0: \theta = 0 \; vs \; H_1: \theta = 1$. Take the strength of the test procedure as $(\alpha = 0.05, \beta = 0.05)$. The normal probability density function is given by

$f(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} exp\left(\frac{-1(x-\mu)^2}{2\sigma^2}\right), -\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0$. $R_m = \frac{P(.|H_1)}{P(.|H_0)}$.

(4)

The boundary points (A, B) for the SPRTP of strength (α, β) are $A = \frac{1-\beta}{\alpha}$ and $B = \frac{\beta}{1-\alpha}$

5A) Based on the layout given below, identify the design. Give proper justification.

Treatment (Catalyst)	**Block (Batch of Raw material)**			
	1	**2**	**3**	**4**
1	73	74	-	71
2	-	75	67	72
3	73	75	68	-
4	75	-	72	75

(3)

5B) Complete the ANOVA table of an RCBD with 4 treatments and 6 blocks.

Source of Variation	Df	SS	MS	F
Treatment		178.17		
Block		192.25		
Error		109.89		
Total	23	480.31		

(3)

5C) The demand for a particular spare part in a factory was found to vary from day to day. In a sample survey, the following information was obtained.

Days	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Number of parts demanded	1124	1125	1110	1120	1126	1115

(4)

Test the hypothesis that the number of parts demanded does not depend on the day of the week. The threshold at 5% level of significance for the required degrees of freedom is 11.07.

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